

Brian H. Hu

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I am a Technical Leader at **Kitware, Inc.** My research interests lie at the intersection of Computer Vision and Natural Language Processing, with a current focus on AI Test and Evaluation (T&E) and AI Alignment. I completed my PhD in Biomedical Engineering from Johns Hopkins University, with a thesis on biologically-inspired artificial intelligence. I have served as the Principal Investigator or Chief Scientist on numerous Department of Defense (DoD)-funded programs, leading teams of AI researchers and engineers to tackle complex problems.

Education

Johns Hopkins University, Baltimore, MD

Ph.D., Biomedical Engineering, October 2017

University of Pittsburgh, Pittsburgh, PA

B.S., Bioengineering (Minor in Chemistry), *summa cum laude*, May 2011

Work Experience

Kitware, Inc., Arlington, VA

Technical Leader, January 2025 - present

Staff Research and Development Engineer, January 2023 - December 2024

Senior Research and Development Engineer, April 2020 - December 2022

- Large language models, alignment techniques
- AI test and evaluation, open source software development
- Explainable AI (XAI), ethical AI, saliency algorithms
- Human-machine teaming, image and video retrieval, anomaly detection
- Small object detection, person/vehicle re-identification

Allen Institute for Brain Science, Seattle, WA

Scientist II, January 2020 - March 2020

Scientist I, December 2017 - December 2019

- Short-term synaptic plasticity and memory in neural networks
- Figure-ground representation in deep neural networks
- Convolutional neural networks with extra-classical receptive fields
- Structure and generalization properties of noise
- Generative adversarial networks for inferring cortical learning rules
- Data-driven, predictive models of neural activity

Skills

Programming: Python, MATLAB, familiar with C/C++, JAVA, LabVIEW

Other Tools: PyTorch, Tensorflow, Jupyter, scikit-learn, pandas, Git, L^AT_EX, HTCondor, NEST, Neuron

Operating Systems: Linux, Windows, Mac OS

Publications

* denotes equal contribution

30. Hashemi, C. , Ravichandran, B. , Anderson, C. , Govil, S. , RichardWebster, B. , **Hu, B.** , and Fieldhouse, K. Improving semantic coverage in ai test and evaluation through generative augmentation. In Harguess, J. D. , Bastian, N. D. , and Pace, T. L. , editors, *Assurance and Security for AI-enabled Systems 2026*, volume 14046, page 366. International Society for Optics and Photonics, SPIE, 2026
29. Menon, N. , Smith, E. , Bastian, N. D. , Hoogs, A. , Basharat, A. , and **Hu, B.** Metacognition ai framework for effective fusion and reasoning over media forensics analytics. In Harguess, J. D. , Bastian, N. D. , and Pace, T. L. ,

- editors, *Assurance and Security for AI-enabled Systems 2026*, volume 14046, page 387. International Society for Optics and Photonics, SPIE, 2026
28. Mujtaba, D. , **Hu, B.** , Hoogs, A. , and Basharat, A. Aligning machiavellian agents: Behavior steering via test-time policy shaping. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 37738–37746, 2026
 27. Adams, J. , **Hu, B.** , Veenhuis, E. , Joy, D. , Ravichandran, B. , Bray, A. , Hoogs, A. , and Basharat, A. Steerable pluralism: Pluralistic alignment via few-shot comparative regression. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, volume 8, pages 15–25, 2025
 26. Ravichandran, B. , Joy, D. , Elliott, P. , **Hu, B.** , Adams, J. , Funk, C. , Veenhuis, E. , Hoogs, A. , and Basharat, A. ALIGN: Prompt-based attribute alignment for reliable, responsible, and personalized LLM-based decision-making. In *ICML 2025 Workshop on Reliable and Responsible Foundation Models*, 2025. URL <https://openreview.net/forum?id=iQptQH12zD>
 25. Lynch, A. , Ravichandran, B. , Webster, B. R. , Veenhuis, E. , Crowell, S. , Collins, R. , Whitesell, A. , Hoogs, A. , and **Hu, B.** Explaining model robustness: combining saliency maps and natural robustness testing using xaitk and nrtk. In Harguess, J. D. , Bastian, N. D. , and Pace, T. L. , editors, *Assurance and Security for AI-enabled Systems 2025*, volume 13476, page 1347603. International Society for Optics and Photonics, SPIE, 2025. doi: 10.1117/12.3052320. URL <https://doi.org/10.1117/12.3052320>
 24. **Hu, B.** , Chan, D. , Sorensen, T. , Chen, X. , Ji, H. , Choi, Y. , Darrell, T. , and Basharat, A. A roadmap for alignable algorithmic decision-makers in the medical triage domain. In *2025 IEEE Conference on Artificial Intelligence (CAI)*, pages 1179–1183. IEEE, 2025
 23. **Hu, B.** , RichardWebster, B. , Tunison, P. , Veenhuis, E. , Ravichandran, B. , Lynch, A. , Crowell, S. , Genova, A. , Bolea, V. , Jourdain, S. , and Whitesell, A. Nrtk: an open source natural robustness toolkit for the evaluation of computer vision models. In Harguess, J. and Bastian, N. , editors, *Assurance and Security for AI-enabled Systems*. International Society for Optics and Photonics, SPIE, 2024
 22. **Hu, B.** , Ray, B. , Leung, A. , Summerville, A. , Joy, D. , Funk, C. , and Basharat, A. Language models are alignable decision-makers: dataset and application to the medical triage domain. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Industry Track, NAACL 2024, Hybrid: Mexico City, Mexico + Online, June 16-21, 2024*. Association for Computational Linguistics, 2024
 21. Lee, J. H. , Choe, Y. , Ardid, S. , Abbasi-Asl, R. , McCarthy, M. , and **Hu, B.** Functional microcircuits in the brain and in artificial intelligent systems. *Frontiers in Computational Neuroscience*, 17:1135507, 2023
 20. **Hu, B.** , Tunison, P. , RichardWebster, B. , and Hoogs, A. Xaitk-saliency: an open source explainable ai toolkit for saliency. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 15760–15766, 2023
 19. Voina, D. , Recanatesi, S. , **Hu, B.** , Shea-Brown, E. , and Mihalas, S. Single circuit in v1 capable of switching contexts during movement using an inhibitory population as a switch. *Neural Computation*, pages 1–54, 2022
 18. RichardWebster*, B. , **Hu***, **B.** , Fieldhouse, K. , and Hoogs, A. Doppelganger saliency: Towards more ethical person re-identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2847–2857, 2022
 17. **Hu, B.** , Vasu, B. , and Hoogs, A. X-MIR: Explainable Medical Image Retrieval. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, January 2022
 16. Schneider-Mizell, C. M. , Bodor, A. L. , ... , **Hu, B.** , ... , Reid, R. C. , and da Costa, N. M. Structure and function of axo-axonic inhibition. *eLife*, 10:e73783, 2021
 15. **Hu, B.** , Tunison, P. , Vasu, B. , Menon, N. , Collins, R. , and Hoogs, A. XAITK: The Explainable AI Toolkit. *Applied AI Letters*, page e40, 2021. doi: <https://doi.org/10.1002/ail2.40>
 14. Vasu, B. , **Hu, B.** , Dong, B. , Collins, R. , and Hoogs, A. Explainable, Interactive Content-Based Image Retrieval. *Applied AI Letters*, page e41, 2021. doi: <https://doi.org/10.1002/ail2.41>
 13. **Hu, B.** , Garrett, M. , Groblewski, P. A. , Ollerenshaw, D. , Shang, J. , Roll, K. , Manavi, S. , Koch, C. , Olsen, S. R. , and Mihalas, S. Adaptation supports short-term memory in a visual change detection task. *PLoS computational biology*, 17(9):e1009246, 2021
 12. Siegle, J. H. , Jia, X. , ... , **Hu, B.** , ... , Olsen, S. R. , and Koch, C. Survey of spiking in the mouse visual system reveals functional hierarchy. *Nature*, pages 1–7, 2021

11. Wagatsuma, N. , **Hu, B.** , von der Heydt, R. , and Niebur, E. Analysis of spiking synchrony in visual cortex reveals distinct types of top-down modulation signals for spatial and object-based attention. *PLoS computational biology*, 17(3):e1008829, 2021
10. Iyer, R. , **Hu, B.** , and Mihalas, S. Contextual integration in cortical and convolutional neural networks. *Frontiers in Computational Neuroscience*, 14:31, 2020. doi: 10.3389/fncom.2020.00031
9. **Hu, B.** , von der Heydt, R. , and Niebur, E. Figure-ground organization in natural scenes: Performance of a recurrent neural model compared with neurons of area v2. *eNeuro*, 6(3), 2019. doi: 10.1523/ENEURO.0479-18.2019
8. **Hu, B.** , Khan, S. , Niebur, E. , and Tripp, B. Figure-ground representation in deep neural networks. In *IEEE CISS-2019 53rd Annual Conference on Information Sciences and Systems*, pages 1–6, Baltimore, MD, 2019. IEEE Information Theory Society. doi: 10.1109/CISS.2019.8693039
7. **Hu***, **B.** , Iyer*, R. , and Mihalas, S. Convolutional neural networks with extra-classical receptive fields. In *Thirty-third Conference on Neural Information Processing Systems (NeurIPS)*, Neuro AI Workshop, Vancouver, Canada, 2019. URL <https://openreview.net/forum?id=rkxSEQtLUS>
6. **Hu, B.** , Shang, J. , Iyer, R. , Siegle, J. , and Mihalas, S. Does the neuronal noise in cortex help generalization? In *Thirty-third Conference on Neural Information Processing Systems (NeurIPS)*, Neuro AI Workshop, Vancouver, Canada, 2019. URL <https://openreview.net/forum?id=S1gc4XF8Lr>
5. **Hu, B.** , Johnson-Bey, I. , Sharma, M. , and Niebur, E. Head movements are correlated with other measures of visual attention at smaller spatial scales. In *IEEE CISS-2018 52nd Annual Conference on Information Sciences and Systems*, pages 1–6, Princeton, NJ, 2018. IEEE Information Theory Society. doi: 10.1109/CISS.2018.8362264
4. **Hu, B.** and Niebur, E. A recurrent neural model for proto-object based contour integration and figure-ground segregation. *Journal of computational neuroscience*, 43(3):227–242, 2017. doi: 10.1007/s10827-017-0659-3
3. **Hu, B.** , Johnson-Bey, I. , Sharma, M. , and Niebur, E. Head movements during visual exploration of natural images in virtual reality. In *IEEE CISS-2017 51st Annual Conference on Information Sciences and Systems*, pages 1–6, Baltimore, MD, 2017. IEEE Information Theory Society. doi: 10.1109/CISS.2017.7926138
2. **Hu, B.** , Kane-Jackson, R. , and Niebur, E. A proto-object based saliency model in three-dimensional space. *Vision Research*, 119:42–49, 2016. doi: 10.1016/j.visres.2015.12.004
1. **Hu, B.** , von der Heydt, R. , and Niebur, E. A neural model for perceptual organization of 3D surfaces. In *IEEE CISS-2015 49th Annual Conference on Information Sciences and Systems*, pages 1–6, Baltimore, MD, 2015. IEEE Information Theory Society. doi: 10.1109/CISS.2015.7086906

Honors and Awards

Honorable Mention, Pytorch Hackathon (Responsible AI), 2021
 Computational and Systems Neuroscience (Cosyne) Travel Award, 2017
 OIST Computational Neuroscience Course (OCNC) Travel Award, 2014
 Visual Neuroscience Training Program Fellowship, 2012–2013
 Neural Engineering Training Initiative Fellowship, 2011–2012
 University Honors College Full Tuition Scholarship, 2007–2011

Professional Service and Activities

Program Chair, ICML '26 Combining Theory and Benchmarks (CTB) Workshop, 2026
 Program Chair and Organizer, AAAI Fall Symposium on AI Trustworthiness and Risk Assessment for Challenged Contexts (ATRACC), 2024–2026
 Area Chair, Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) Industry Track, 2025
 Program Chair, AAAI Fall Symposium on Assured and Trustworthy Human-centered AI (ATHAI), 2023
 Reviewer for *Journal of Cognitive Neuroscience*, *Frontiers in Systems Neuroscience*, *IEEE Transactions on Image Processing*, *CVPR*, *ICCV*, *WACV*, *Neurips*